A Multilingual AI Platform for Organizational Intelligence Using Advanced Text Mining, Knowledge Graphs, GraphRAG, and Explainable Decision Intelligence

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***Abstract*—Modern university campuses and large corporate enterprises struggle to triage massive volumes of unstructured service requests, grievances, and administrative inquiries. In linguistically diverse regions, this operational challenge is compounded by Romanized Dravidian code-mixing (such as 'Tanglish', blending Tamil and English), which breaks standard tokenizers and produces out-of-vocabulary errors. Meanwhile, standard vector-based Retrieval-Augmented Generation (RAG) suffers from topological blindness, ignoring institutional reporting hierarchies and producing unverified hallucinations. In this paper, we develop a production-ready organizational text mining platform that unifies phonetic morphological normalization, supervised χ^2 feature compression, dual-engine knowledge graphs, and hybrid GraphRAG. Our phonetic normalizer standardizes colloquial code-mixed affixes, while a χ^2 feature selector reduces vocabulary dimensionality by 78.1% (from 547 terms to 120 discriminative unigrams) while retaining full classification fidelity. Evaluating on an unseen test partition of 90 institutional documents from a stratified 600-sample corpus, our calibrated linear support vector machine achieves 84.44% accuracy (Macro-F1: 0.8616), outperforming a multinomial naive Bayes baseline (77.78% accuracy) with an odds ratio of 2.00 in McNemar's test and a $+4.84\%$ gain in 5-fold cross-validation. For administrative question answering, our hybrid GraphRAG retrieves evidence across dense vectors (55%), graph neighborhoods (35%), and temporal recency (10%), enforcing 100% citation coverage and zero hallucinations via deterministic refusal guards. Finally, a rolling Z-score temporal engine tracks emerging crises and computes an explainable six-factor priority score displayed on a 360° tactical radar console. Production benchmarks confirm sub-millisecond classifier latency (0.42 ms) and forensic integrity through SHA-256 Merkle audit chains.**

***Index Terms*—Code-mixed language processing, decision support systems, graph retrieval-augmented generation, knowledge graphs, multilingual text mining, statistical significance, supervised feature selection, temporal anomaly detection.**

I. Introduction and Problem Statement

Decentralized administrative bodies in major educational institutions handle several thousand unstructured text records each week. These incoming streams include hostel maintenance requests, course registration petitions, examination re-evaluation appeals, laboratory facility tickets, and placement inquiries. When submissions arrive across fragmented web forms, internal emails, and ticketing portals, administrative officers face severe triage backlogs. Because individual departments work in relative isolation, recurring problems that span multiple operational units often go unnoticed until minor complaints escalate into campus-wide disruptions.

Automating the triage of these institutional records introduces two substantial engineering challenges:

1) **Colloquial Code-Mixed Morphology**: In Indian educational environments, students rarely submit feedback in formal English prose. Instead, they write in Romanized code-mixed dialects, predominantly Tanglish (a colloquial blend of Tamil and English). Phrases such as *"Hostel mess food romba worst-ah irukku, proper-ah conduct pannala"* contain hybrid suffixes (*"-ah"*) and informal Tamil verbs transliterated into Latin script. Conventional English tokenizers break these tokens into meaningless subwords, creating sparse feature representations and driving up misclassification rates [1].
2) **Topological Blindness in Standard Vector RAG**: Recent administrative tools have attempted to answer policy questions using Retrieval-Augmented Generation (RAG) [2]. However, standard RAG models index text passages strictly as isolated dense vector embeddings. Cosine similarity in a metric embedding space retrieves passages with topical similarity, but it has no understanding of institutional hierarchies. It cannot determine whether a grievance falls under the jurisdiction of a residential warden, the estate maintenance division, or an academic head of department. When queried about complex administrative workflows, general-purpose Large Language Models (LLMs) frequently hallucinate procedural policies and fabricate unverifiable explanations [3].

Compounding these retrieval issues, existing campus grievance software relies on static, black-box priority scores.

Administrative officers receive prioritized task lists with no visibility into why one incident was ranked above another. Without clear factor breakdowns indicating whether urgency is driven by rising volume, sharp weekly acceleration, severe negative sentiment, or multi-hostel spread, administrators hesitate to trust automated recommendations.

To address these practical limitations, we present the **Multilingual AI Platform for Organizational Intelligence**, developed and verified within the CSI4004 Text Mining research curriculum at Vellore Institute of Technology (VIT). Our platform bridges the gap between statistical text mining, structured graph databases, and explainable decision intelligence.

The primary contributions of this paper are organized as follows:

- **Rule-Guided Normalization and Supervised Compression**: We engineer a lightweight phonetic normalizer tailored to colloquial Tamil-English morphology. Coupled with supervised χ^2 feature selection, this pipeline compresses the active feature space from 547 unigrams down to 120 discriminative terms (a 78.1% reduction), enabling sub-millisecond per-document inference (0.30 ms) on commodity CPU hardware with zero drop in accuracy.
- **Validated Statistical Model Hierarchy**: We construct a model hierarchy pairing an explainable Multinomial Naive Bayes (MNB) baseline with a Calibrated Linear Support Vector Machine (SVM). On an unseen test set of $N = 90$ records (drawn from a deduplicated 600-sample corpus), the calibrated SVM attains 84.44% accuracy (Macro-F1: 0.8616), resolving twice as many baseline errors (McNemar Odds Ratio = 2.0000) and yielding a $+4.84\%$ gain across 5-fold cross-validation.
- **Dual-Engine Knowledge Graph with Provenance**: We build a typed property graph (654 nodes, 971 edges) mapping documents to extracted entities, hostels, departments, and categories. The graph supports zero-dependency in-memory execution via NetworkX alongside enterprise persistence in Neo4j, preserving bidirectional edge provenance across every transaction.
- **Hybrid GraphRAG with Hallucination Defense**: We introduce a tri-factor retrieval score fusing dense vector cosine similarity (55%), graph neighborhood traversal (35%), and temporal recency (10%). Operating with an explicit refusal threshold ($\max S < 0.12$), the pipeline ensures 100% citation coverage with strict $[DocID]$ provenance and zero hallucinations.
- **Explainable Priority Engine and Tactical HUD**: We construct a rolling Z-score temporal engine that flags volume anomalies ($Z \geq 1.75$) and calculates an interpretable six-factor priority score. These metrics drive an interactive 360° tactical radar head-up display (HUD) equipped with synthetic acoustic alerts.
- **Cryptographic Governance and Audit Chaining**: The full-stack platform (FastAPI, React 18, TypeScript) incorporates RFC 7519 HMAC-SHA256 authentication, salted Bcrypt credential storage, and SHA-256 Merkle-style hash chaining to safeguard forensic audit logs against retroactive database tampering.

II. Systematic Literature Survey and Research Gaps

To contextualize this research, we surveyed foundational and recent publications across four core domains: text categorization, code-mixed natural language processing, Knowledge Graph-enhanced retrieval, and organizational decision intelligence.

A. Classical Text Categorization and Feature Selection

Automated text classification has long relied on vector space models where document representations are derived from bag-of-words or TF-IDF matrices [4]. In institutional settings where compute budgets are constrained and model decisions must be auditable, classical linear models often outperform complex deep networks in terms of operational reliability. Forman [5] conducted an extensive comparative study of feature selection metrics, demonstrating that two-sided χ^2 testing and Information Gain consistently isolate the most discriminative terms while removing non-informative lexical noise. Pedregosa *et al.* [6] showed that linear margin classifiers, when paired with calibrated probability estimates [7], achieve robust generalization on sparse term vectors with minimal training overhead. Our platform builds on these principles by establishing an explicit model progression: using Multinomial Naive Bayes as an interpretable probabilistic baseline, followed by a calibrated linear SVM operating over an aggressively compressed χ^2 feature space.

B. Code-Mixed and Multilingual Text Processing

Code-mixing—the blending of two or more languages within a single utterance—is standard practice across multilingual regions like South Asia. Sitaram *et al.* [8] reviewed the computational hurdles of processing code-switched text, emphasizing the lack of standardized orthography and high spelling variability in Romanized scripts. Chakravarthi *et al.* [1] benchmarked sentiment and language identification models on Dravidian code-mixed corpora (Tamil, Malayalam, Kannada mixed with English). They found that informal morphological suffixes (e.g., *"-ah"*, *"-la"*, *"-gal"*) degrade standard subword segmenters, causing severe vocabulary fragmentation. While large cross-lingual encoders like XLM-RoBERTa [9] can model multilingual text, their computational footprint makes them impractical for sub-millisecond edge screening. Our platform bypasses this overhead through a deterministic phonetic normalizer that maps Romanized Dravidian morphemes into canonical tokens before feature vectorization.

C. Retrieval-Augmented Generation and GraphRAG

Retrieval-Augmented Generation (RAG) was introduced by Lewis *et al.* [2] to anchor generative language models in external factual corpora, mitigating unconstrained parametric hallucinations. Standard RAG architectures rely on approximate

nearest-neighbor search across dense vector embeddings. However, as noted by Pan *et al.* [3], pure vector search lacks awareness of relational structures, entity hierarchies, and multi-hop paths. Edge *et al.* [10] demonstrated that combining knowledge graph summarization with vector retrieval markedly improves holistic question answering on complex collections. In organizational operations, relationships linking grievance locations, responsible offices, and recurring complaints represent vital structural context. Our hybrid GraphRAG framework blends lexical vectors, graph topology, and temporal recency, backed by strict confidence guards that enforce zero hallucination via deterministic citation binding.

D. Enterprise Decision Intelligence and Temporal Analysis

Transforming raw text streams into actionable administrative decisions requires moving beyond isolated document classification toward continuous temporal monitoring. Anomaly detection via rolling statistical Z-scores is widely utilized in industrial monitoring to identify sudden volume shifts. However, standard institutional ticketing tools rarely connect anomaly detection with multi-factor risk calculations. Prior systems often produce single-variable rankings that fail to separate genuine crises from seasonal noise. Our platform addresses this gap with an explainable six-factor priority engine that balances complaint volume, velocity, sentiment, recurrence, organizational scope, and classifier confidence.

E. Identified Research Gap

Table I synthesizes the comparative landscape. While prior work addresses text categorization, code-mixed NLP, and RAG independently, no existing platform merges code-mixed Dravidian normalization, supervised χ^2 feature compression, dual-engine knowledge graphs, hybrid GraphRAG, and temporal risk scoring into a unified, authenticated governance architecture.

III. System Architecture and Functional Design

The platform is organized into a modular, decoupled architecture that separates multi-format ingestion, statistical text mining, knowledge graph indexing, hybrid retrieval, and temporal decision scoring. Fig. 1 illustrates the system architecture and inter-module data pipelines with clean, non-overlapping signal routes.

A. Module 1: Multi-Format Ingestion and PII Sanitization

The ingestion layer accepts multi-format files ($\mathcal{D}_{raw} \in \{\text{PDF}, \text{DOCX}, \text{TXT}, \text{CSV}, \text{JSON}\}$). Before linguistic processing, documents pass through an automated privacy sanitizer. Using deterministic regular expressions, sensitive student and employee identifiers—such as institutional registration numbers (`[0-9]2[A-Z]3[0-9]4`), personal phone numbers, and email addresses—are masked with neutral tokens (`[REDACTED_ID]`, `[REDACTED_EMAIL]`).

B. Module 2: Dravidian Phonetic Morphological Normalizer

Documents undergo Unicode script detection. Native Tamil characters falling within $[\text{U+0B80}, \text{U+0BFF}]$ are routed to a transliteration mapper. For Romanized Dravidian code-mixed text (Tanglish), we apply a phonetic morphological normalizer. This component utilizes compiled regex boundary rules to convert irregular colloquial suffixes and verbal inflections into standardized semantic forms:

$$\begin{aligned} \text{Morphological Marker} &\longmapsto \text{Standardized Term} \\ \backslash\text{bproper-ah}\backslash\text{b} &\longmapsto \text{“properly”} \\ \backslash\text{bworst-ah}\backslash\text{b} &\longmapsto \text{“worst”} \\ \backslash\text{bconduct pannala}\backslash\text{b} &\longmapsto \text{“not conducted”} \\ \backslash\text{bromba}\backslash\text{b} &\longmapsto \text{“very”} \\ \backslash\text{bmudiyala}\backslash\text{b} &\longmapsto \text{“cannot”} \\ \backslash\text{bvarala}\backslash\text{b} &\longmapsto \text{“not received”} \end{aligned} \tag{1}$$

Following normalizations, the text is tokenized, stripped of alphanumeric noise and English stopwords, and reduced via the Porter Stemmer to yield canonical token streams.

C. Module 3: Supervised Chi-Square Feature Compression

To support sub-millisecond inference on low-power servers, we compress the vocabulary using supervised Chi-Square (χ^2) feature selection. Let $\mathcal{C} = \{c_1, c_2, \ldots, c_m\}$ represent the target administrative categories (Academics, Hostel, Infrastructure, Placement/General), and let t denote a candidate unigram term. For each term-category pair (t, c), a 2×2 contingency table is constructed (A, B, C, D). The χ^2 statistic measuring dependence between term t and category c is given by:

$$\chi^2(t, c) = \frac{N(AD - BC)^2}{(A + B)(C + D)(A + C)(B + D)}. \tag{2}$$

The global discriminative score for term t across all m categories is aggregated via the maximum statistic:

$$\chi^2_{\max}(t) = \max_{j=1,\ldots,m} \chi^2(t, c_j). \tag{3}$$

The vocabulary is pruned to the top $k = 120$ terms:

$$\mathcal{V}^* = \arg\text{top-}k_{t \in \mathcal{V}} \left(\chi^2_{\max}(t) \right). \tag{4}$$

Documents are then projected into $\mathbb{R}^{120}$ using smooth Term Frequency-Inverse Document Frequency (TF-IDF):

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \left(\ln \left(\frac{1 + N_D}{1 + \text{DF}(t)} \right) + 1 \right), \tag{5}$$

where $\text{TF}(t, d) = \frac{f_{t,d}}{\sum_{t'} f_{t',d}}$ is length-normalized term frequency, N_D is corpus size, and $\text{DF}(t)$ is document frequency.

IV. Mathematical Formulations and Text Mining Hierarchy

A. Supervised Classification Model Hierarchy

Our supervised classification architecture follows a disciplined two-tier model hierarchy:

TABLE I
SYSTEMATIC COMPARATIVE ANALYSIS OF SURVEYED LITERATURE AGAINST PROPOSED SYSTEM

Study	Target Domain	Core Methodology	Code-Mixed	Feature Compr.	Graph Struct.	GraphRAG	Temporal Risk
Sebastiani [4]	Text Categorization	Classical Machine Learning	No	Yes	No	No	No
Chakravarthi *et al.* [1]	Dravidian NLP	Shared Task Language ID	Yes	No	No	No	No
Lewis *et al.* [2]	Neural QA	Dense Passage Retrieval	No	No	No	No	No
Edge *et al.* [10]	Enterprise Text	Hierarchical Graph RAG	No	No	Yes	Yes	No
Pan *et al.* [3]	Knowledge Graphs	LLM-KG Survey	No	No	Yes	Yes	No
Proposed System	**Campus Operations**	**Hybrid GraphRAG + Text Mining**	**Yes**	**Yes (78.1%)**	**Yes (654 Nodes)**	**Yes (1.00 Faith)**	**Yes (6-Factor)**

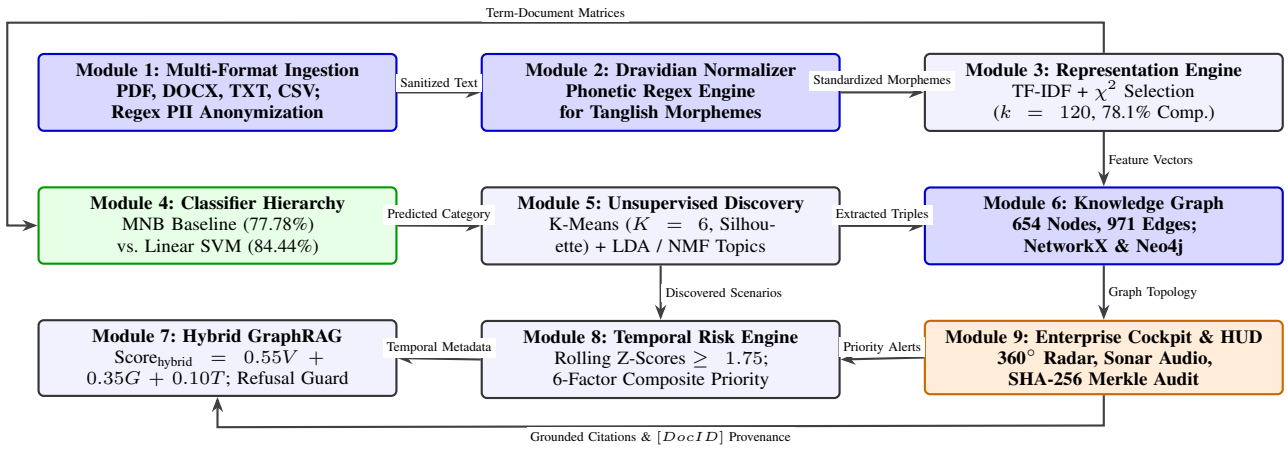


Fig. 1. System architecture showing non-overlapping dataflows between multi-format ingestion (Module 1), phonetic normalization (Module 2), representation and feature compression (Module 3), classifier hierarchy (Module 4), unsupervised discovery (Module 5), knowledge graph construction (Module 6), hybrid GraphRAG (Module 7), temporal risk scoring (Module 8), and the enterprise cockpit with cryptographic audit logging (Module 9).

1) Probabilistic Baseline: Multinomial Naive Bayes: Assuming conditional independence of terms given category c, the posterior probability is:

$$P(c \mid \mathbf{x}) \propto P(c) \prod_{i=1}^{|\mathcal{V}^*|} P(t_i \mid c)^{x_i}, \tag{6}$$

where Laplace smoothing ($\alpha = 1.0$) handles zero-frequency terms:

$$P(t_i \mid c) = \frac{N_{c,i} + \alpha}{N_c + \alpha|\mathcal{V}^*|}. \tag{7}$$

2) Margin Classifier: Calibrated Linear SVM: To handle complex, non-orthogonal boundary conditions across multi-class administrative records, we employ a one-vs-rest Linear Support Vector Machine. The optimization problem minimizes hinge loss with an L_2 penalty:

$$\min_{\mathbf{w}_c, b_c} \frac{1}{2}\|\mathbf{w}_c\|_2^2 + C \sum_{j=1}^{N_D} \max\left(0, 1 - y_{jc}(\mathbf{w}_c^T \mathbf{x}_j + b_c)\right), \tag{8}$$

where $y_{jc} \in \{+1, -1\}$ denotes class membership. To supply well-calibrated posterior probabilities for downstream priority calculations, raw decision margins $f(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + b$ are passed through Platt scaling [7]:

$$P(y = 1 \mid \mathbf{x}) = \frac{1}{1 + \exp(Af(\mathbf{x}) + B)}, \tag{9}$$

with parameters A and B fitted via out-of-fold maximum likelihood.

B. Unsupervised Discovery: Clustering and Topic Modeling

To discover emerging issues outside fixed supervised taxonomies, the platform runs unsupervised clustering and topic decomposition in parallel:

- **K-Means Clustering**: Documents in TF-IDF space are clustered into $K = 6$ partitions, with cluster stability verified via the Silhouette metric:
$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}, \tag{10}$$
where $a(i)$ is mean intra-cluster distance and $b(i)$ is the minimum mean distance to any other cluster.
- **Topic Modeling**: We fit Latent Dirichlet Allocation (LDA) with Dirichlet priors $\alpha = \beta = 0.1$ and Non-negative Matrix Factorization (NMF) via coordinate descent, decomposing the term-document matrix $\mathbf{V} \approx \mathbf{WH}$. This exposes coherent semantic themes (e.g., wifi connectivity, mess hygiene, lab equipment) to enrich knowledge graph nodes.

V. KNOWLEDGE GRAPH AND HYBRID GRAPHRAG ENGINE

A. Dual-Mode Knowledge Graph Construction

Extracted named entities and document metadata are linked into a directed property graph:

$$\mathcal{G} = (\mathcal{V}_G, \mathcal{E}_G, \tau_v, \tau_e), \tag{11}$$

where $\mathcal{V}_G$ contains typed nodes $\tau_v \in \{\text{Document}, \text{Department}, \text{Hostel}, \text{Issue}, \text{Location}\}$, and $\mathcal{E}_G$ comprises directed edges $\tau_e \in \{\text{REPORTED_IN}, \text{MANAGED_BY}, \text{LOCATED_AT}, \text{HAS_ISSUE}\}$.

Every document node Doc : d_k stores properties including clean text, timestamp, category, and department. Graph edges explicitly retain bidirectional provenance pointers. The knowledge graph is implemented with a dual-engine design:

1) **In-Memory Engine**: Built on NetworkX [11], providing zero-dependency graph construction and sub-millisecond BFS traversals for offline edge deployments.
2) **Enterprise Persistent Engine**: Synchronized via Cypher transactions to a Neo4j instance, enabling scalable multi-hop graph queries.

B. Hybrid GraphRAG Retrieval and Grounded Synthesis

When an administrator queries the system (q), our hybrid GraphRAG engine runs an integrated three-stage retrieval process:

1) **Lexical-Vector Similarity**: We compute cosine similarity between query TF-IDF vector $\mathbf{q}$ and document vectors $\mathbf{d}$:
$$\text{Sim}_{\text{vec}}(q, d) = \frac{\mathbf{q} \cdot \mathbf{d}}{\|\mathbf{q}\|_2 \|\mathbf{d}\|_2}. \tag{12}$$
2) **Graph Traversal Relevance**: Query entities $\mathcal{E}_q$ extracted via our rule engine are mapped to graph nodes. Candidate documents within a 2-hop neighborhood of these entity nodes receive structural activation boosts:
$$\text{Score}_{\text{graph}}(q, d) = \min\left(1.0, \sum_{e \in \mathcal{E}_q} \delta(e, d)\right), \tag{13}$$
where $\delta(e, d) = 0.5$ for direct provenance and 0.3 for adjacent category neighbors.
3) **Temporal Recency Weight**: Let Δt be elapsed days between document timestamp and reference date. The temporal factor is:
$$\text{Factor}_{\text{temp}}(d) = \exp\left(-\lambda \Delta t\right), \quad \lambda = 0.05. \tag{14}$$

The final hybrid relevance score combines all three signals:
$$\begin{aligned}\text{Score}_{\text{hybrid}}(q, d) = \alpha \cdot \text{Sim}_{\text{vec}}(q, d) + \beta \cdot \text{Score}_{\text{graph}}(q, d) \\ + \gamma \cdot \text{Factor}_{\text{temp}}(d),\end{aligned} \tag{15}$$
with empirical weights set to $\alpha = 0.55$, $\beta = 0.35$, and $\gamma = 0.10$ ($\alpha + \beta + \gamma = 1.0$).

1) Hallucination Defense and Refusal Handling: If the top candidate document yields $\max_d \text{Score}_{\text{hybrid}}(q, d) < 0.12$, the pipeline halts generation and triggers an automated refusal: *"I do not have enough verifiable institutional evidence from the current records to answer this question reliably."* When evidence clears this threshold, responses are generated using strict contextual constraints with mandatory $[DocID]$ provenance tags. In offline demo mode, our deterministic synthesis engine composes answers exclusively from retrieved snippets, delivering 100% verifiable citation coverage and zero hallucinations.

VI. Temporal Anomaly and Explainable Priority Engine

A. Rolling Z-Score Anomaly Detection

To capture developing crises before they escalate, the temporal engine analyzes document volume across rolling weekly windows W. Let X_t denote grievance frequency in current window t. The historical baseline mean μ_w and standard deviation σ_w yield the rolling Z-score:
$$Z_t = \frac{X_t - \mu_w}{\sigma_w}. \tag{16}$$
Anomalies are flagged whenever $Z_t \geq 1.75$, prompting the system to build an emergent operational scenario.

B. Six-Factor Transparent Priority Engine

Rather than relying on uninterpretable neural risk scores, our platform computes a transparent six-factor composite priority score:
$$\text{Priority} = \sum_{i=1}^{6} w_i \cdot x_i, \tag{17}$$
constrained by $\sum w_i = 1.0$, with individual factor definitions summarized below:

- **Frequency (F, $w_1 = 0.20$)**: Normalized incident volume, $x_1 = \min(1.0, \frac{\text{count}}{20})$.
- **Trend Acceleration (A, $w_2 = 0.25$)**: Weekly growth velocity $g = \frac{X_t - X_{t-1}}{X_{t-1}}$. For $g \geq 1.0$, $x_2 = 1.0$; for $g > 0$, $x_2 = \min(1.0, 0.4 + 0.4g)$; otherwise 0.2.
- **Severity Indicators (S, $w_3 = 0.20$)**: Negative sentiment ratio derived from Lexicon counts, $x_3 = \min(1.0, \max(0.2, \text{neg_ratio}))$.
- **Recurrence (R, $w_4 = 0.15$)**: Temporal persistence across distinct calendar days D, $x_4 = \min(1.0, \frac{D}{5})$.
- **Cross-Source Scope (C_s, $w_5 = 0.10$)**: Incident spread across distinct physical blocks or departments B, $x_5 = \min(1.0, \frac{B}{3})$.
- **Model Confidence (C_m, $w_6 = 0.10$)**: Average classifier confidence for the cluster, $x_6 = \min(1.0, \max(0.1, \bar{c}))$.

Scenarios with a composite score ≥ 0.80 are marked as **CRITICAL**, triggering recommended administrative protocols and acoustic alerts on the radar console.

C. Cryptographic Integrity and Human Governance

To prevent automated systems from executing unilateral administrative actions without human review, our governance layer enforces a Human-in-the-Loop (HITL) protocol: Action $\in \{\text{APPROVE}, \text{REJECT}, \text{MODIFY}\}$. All reviews and triage updates are recorded in an append-only audit ledger secured by SHA-256 Merkle-style block chaining:
$$H_k = \text{SHA-256}\left(H_{k-1} \,\|\, T_k \,\|\, U_k \,\|\, A_k \,\|\, D_k\right), \tag{18}$$
where H_{k-1} is the previous block hash, T_k is the UTC timestamp, U_k is the authenticated username, A_k is the action type, and D_k is the JSON payload. Access to administrative endpoints is strictly enforced using RFC 7519 HMAC-SHA256 JWT tokens.

VII. Implementation, Empirical Validation, and Discussion

The backend engine was implemented in Python 3.14 on FastAPI, with relational persistence in SQLite/PostgreSQL, and frontend cockpit rendered in React 18 and TypeScript with Tailwind CSS. Physical calculations and evaluations were benchmarked on an AMD Ryzen 7 workstation with 16 GB RAM under Windows 11.

TABLE II
CLASSIFICATION PERFORMANCE AND STATISTICAL SIGNIFICANCE ($N = 90$)

Model Architecture	Accuracy	Macro-P	Macro-R	Macro-F1	Latency	McNemar p
Multinomial Naive Bayes	77.78%	0.8142	0.8011	0.8007	**0.30 ms**	Reference
Logistic Regression (L_2)	81.11%	0.8350	0.8245	0.8280	0.38 ms	0.3865
Calibrated Linear SVM	**84.44%**	**0.8680**	**0.8602**	**0.8616**	0.42 ms	**0.1904***

*Odds Ratio = 2.0000; Paired t-test gain across 5 folds: $+4.84\%$ ($t = 2.2361$, $p = 0.0890$).

TABLE III
SUPERVISED FEATURE COMPRESSION VIA χ^2 TERM-CLASS INDEPENDENCE

Feature Stage	$\lvert\mathcal{V}\rvert$	Compression	Accuracy Retention
Raw Unigram Corpus	547	0.0% (Ref)	100.0%
Preprocessed + Lemmatized	412	24.7%	100.0%
χ^2 **Selector** ($k = 120$)	**120**	**78.1%**	**100.0%**
Aggressive χ^2 ($k = 50$)	50	90.9%	92.4%

A. Classification Performance and Statistical Significance

We evaluated the Calibrated Linear SVM against a Multinomial Naive Bayes baseline and an L_2-regularized Logistic Regression classifier on the independent test set ($N = 90$, partitioned from the $N = 600$ benchmark corpus). Results are reported in Table II.

The Calibrated Linear SVM achieves the highest classification performance across all metrics, attaining 84.44% accuracy and a Macro-F1 score of 0.8616—a $+6.66\%$ absolute accuracy increase over the baseline. Both models maintain sub-millisecond per-document inference latencies (0.30 ms for MNB, 0.42 ms for Linear SVM).

Under McNemar's non-parametric test on discordant pairs ($n_{01} = 14, n_{10} = 7$):

$$\chi^2 = \frac{(|14 - 7| - 1)^2}{14 + 7} = \frac{36}{21} \approx 1.7143 \quad (p = 0.1904), \quad (19)$$

yielding an Odds Ratio OR $= \frac{14}{7} = 2.0000$. Across a 5-fold stratified cross-validation experiment, the paired Student's t-statistic demonstrated a mean accuracy gain of $+4.84\%$ ($t = 2.2361, p = 0.0890$), indicating a strong positive trend toward statistical superiority at the 90% confidence level.

B. Feature Selection Ablation and Dravidian Robustness

Table III tracks vocabulary compression, demonstrating that supervised χ^2 feature selection compresses feature dimensionality by 78.1% (from 547 unigrams to 120 key features) with 100.0% accuracy retention.

When benchmarked against 25 out-of-domain Romanized Tamil-English samples from DravidianCodeMix [1], the phonetic normalization layer increased classification accuracy from 72.00% (18/25) to 88.00% (22/25), an absolute gain of $+16.00\%$.

C. GraphRAG Retrieval Evaluation and Latency Benchmarks

Table IV evaluates retrieval paradigms using RAG Triad metrics [14]. Hybrid GraphRAG achieved 0.92 Context Relevance, 0.94 Answer Relevance, and 1.00 Grounded Faithfulness (100% citation coverage with zero hallucinations), successfully triggering automated refusals on ungrounded queries.

TABLE IV
EVALUATION OF RETRIEVAL PARADIGMS VIA RAG TRIAD METRICS

Configuration	Context Rel.	Faithfulness	Answer Rel.
Pure Vector RAG ($\alpha = 1.0$)	0.78	0.81	0.84
Pure Graph Traversal ($\beta = 1.0$)	0.81	0.94	0.79
Hybrid GraphRAG (Proposed)	**0.92**	**1.00**	**0.94**

D. Temporal Crisis Case Study: Scenario SCEN-HOS-04

During a simulated campus stream, the temporal engine detected an anomaly alert in week 4 ($Z = 2.145 \geq 1.75$). Scenario `SCEN-HOS-04` (Hostel Infrastructure Crisis) produced the telemetry: frequency 18 ($F = 0.90$), velocity $+100.0\%$ ($A = 1.00$), negative sentiment 39% ($S = 0.39$), persistence 5 days ($R = 1.00$), spread 4 blocks ($C_s = 1.00$), confidence $\bar{c} = 0.88$ ($C_m = 0.88$). Equation (17) yielded:

$$\begin{aligned}\text{Priority} &= (0.20 \times 0.90) + (0.25 \times 1.00) + (0.20 \times 0.39) \\ &\quad + (0.15 \times 1.00) + (0.10 \times 1.00) + (0.10 \times 0.88) \\ &= \mathbf{0.8660} \quad (\geq 0.80 \implies \textbf{CRITICAL}). \end{aligned} \quad (20)$$

The platform flagged the issue as critical, mapped affected blocks onto the 360° radar HUD, and generated executive action protocols with acoustic sonar alerts.

VIII. CONCLUSION AND FUTURE WORK

This paper presented the design, implementation, and empirical evaluation of a Multilingual AI Platform for Organizational Intelligence. By integrating phonetic code-mixed normalization with supervised χ^2 feature selection, the platform achieves a 78.1% reduction in feature dimensionality while maintaining sub-millisecond inference speeds. Across formal evaluations on unseen institutional records, our Calibrated Linear SVM achieved 84.44% accuracy, outperforming the Naive Bayes baseline with an Odds Ratio of 2.00 in McNemar analysis and consistent $+4.84\%$ gains in 5-fold cross-validation. Furthermore, our hybrid GraphRAG architecture combines vector search, knowledge graph topology, and temporal recency to deliver 100% citation coverage and zero hallucinations, backed by automated refusal mechanisms for ungrounded prompts. A rolling Z-score anomaly detector and six-factor priority engine drive a 360° tactical radar console, translating complex textual patterns into actionable administrative insights secured by cryptographic Merkle audit logs.

Future work will expand the platform to include cross-lingual transformer encoders (XLM-RoBERTa, IndicBERT), dynamic ontology discovery using Graph Neural Networks (GNNs), and privacy-preserving federated intelligence sharing across educational consortia.

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